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6 June 1966
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4 March 1964

MILITARY SPECIFICATION

BATTERY MARK 46 MOD 1

PRIMARY TYPE, SILVER-ZINC-ALKALI

This specification has been approved by the
Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requirements for the design and procurement of a primary type, silver-zinc-alkali battery, BUWEPS LD 495965, which when assembled within a torpedo provides electric power for its propulsion and control.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, For Packaging
PPP-C-843	Cushioning Material, Cellulosic

MILITARY

MIL-W-76	Wire and Cable, Hookup, Electrical, Insulated
MIL-B-117	Bags and Sleeves, Interior Packaging
MIL-S-901	Shockproof Equipment, Class HI (High-Impact) Shipboard Application, Tests for

FSC 6135

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MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-P-17667	Paper, Wrapping, Chemically Neutral (Noncorrosive)

STANDARDS

MILITARY

MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 495852	Box Assembly
BUWEPS LD 495965	Primary Battery Mk 46 Mod 1
BUWEPS LD 495966	Preserved (Packaging) Condition
BUWEPS LD 495967	Shipping (Packing) Condition
BUWEPS Drawing 1803669	Battery Assembly Mk 46 Mod 1

PUBLICATIONS

Naval Ordnance Laboratory White Oak, Maryland Report NOLTR 64-214	A Method for the Quantitative Analysis of the Silver Oxide Cathode

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20360.)

3. REQUIREMENTS

3.1 Description. Battery Mk 46 Mod 1 is a reserve electrolyte battery of the primary type supplied with all cells in a dry and charged state and requiring only the addition of electrolyte for activation. Electrolyte, of a type and in an amount sufficient to properly activate the battery, is installed in a sealed compartment within the unit at the time of manufacture and is automatically supplied to all cells by means of an electrically triggered activation system. The battery is constructed of two numerically equal, series-connected cell groups having external terminals. With the two cell groups connected in series, the battery will supply a load of 450 amperes at a nominal voltage of 76 volts; parallel connected, the battery will supply a load of 250 amperes at a nominal voltage of 42 volts. In both operating conditions, a separate cell group must supply an additional load of 35 amperes at a nominal voltage of 26 volts.

3.2 General requirements

3.2.1 Contractor-furnished data. Data required by this specification, or by applicable documents referenced in Section 2, need not be furnished unless specified in the contract or order (see 6.2).

3.2.2 Sampling

3.2.2.1 Preproduction sample. Unless otherwise specified in the contract or order, two preproduction samples of Battery Mk 46 Mod 1 shall be manufactured using the same methods and procedures proposed for the production lot. The samples shall be selected and tested as described in Section 4 herein and are for the purpose of determining that the contractor, prior to starting production, is capable of yielding items that comply with the technical requirements of the contract.

3.2.2.2 Periodic production samples. Unless otherwise specified in the contract or order, samples shall be selected and tested as described in 4.3.3 in order to determine continued conformance to the requirements of this specification. In the event periodic production samples fail to conform to specification requirements, the contractor shall be notified of test results.

3.2.2.3 All preproduction and periodic production sample cells shall be delivered as complete cells, without electrolyte, at battery subassembly level.

3.2.3 Materials. All materials and purchased parts entering into the final assembly of the battery shall conform to applicable Federal and Military specifications and BUWEPS LD 495965. Materials which are not

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covered by specifications or which are not specifically designated herein shall be of the best quality, or the lightest weight practicable consistent with strength and durability, and entirely suited for the purpose for which intended under all conditions of test and service operation. Recommendations concerning substitution for any part or material specified herein shall be accompanied by a description of the substitute together with evidence to substantiate any claim as to its suitability. At the discretion of the Naval Ordnance Systems Command, test samples may be required to prove the suitability of the proposed substitute.

3.2.3.1 Government furnished material. When the Government elects to furnish silver, the total quantity shall be based on the total number of complete batteries and sample cells required to be delivered, and for a battery containing a minimum of 700 troy ounces of silver, consisting of 18 auxiliary and 56 propulsion cells.

3.2.3.2 Wire and terminals. Wire and cable, when used, shall conform to Specification MIL-W-76 or Specification MIL-W-16878. Terminals shall be in accordance with BUWEPS LD 495965.

3.2.4 Disposition of cells used in destructive testing. Silver in cells selected in 4.3.1.1 for destructive testing in accordance with MIL-STD-414 shall be provided by the contractor and shall remain his property.

3.2.5 Finishes, coatings, and dimensions. Finishes, coatings, and dimensions for the battery shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUWEPS LD 495965.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BUWEPS LD 495965 and the following special marking:

3.2.6.1 Each battery shall be permanently and legibly marked in a conspicuous place with the following information:

(a) BATTERY MK 46 MOD 1
(b) LD 495965 DWG 1803669
(c) CONTRACT OR ORDER NUMBER
(d) DATE OF MANUFACTURE (MONTH AND YEAR)
(e) MANUFACTURER'S NAME
(f) SERIAL NUMBER

3.2.6.2 Each battery shall bear a distinguishing serial number which will be assigned, upon request, by the Naval Underwater Weapons Research and Engineering Station, Newport, R. I., or as specified in the contract, requisition, or order.

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3.2.6.3 Each battery shall have a large lettered caution notice marked on the outside casing of the battery assembly, reading "CAUTION - READ MANUFACTURER'S INSTRUCTION BEFORE PLACING THIS BATTERY IN SERVICE."

3.2.6.4 The test cells (4.3.1.3) shall be identified by legibly marking each cell with the highest and lowest serial numbers of the production lot of the batteries they represent.

3.2.6.5 Terminals shall be marked as shown on the applicable drawings of BUWEPS LD 495965.

3.2.7 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.8 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified in the applicable drawings.

3.2.9 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein, shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2.10 General instructions

3.2.10.1 Manufacturer's instructions. Unless otherwise specified by the procuring agency, each battery shall be accompanied by a set of special instructions for installing the battery and for performing any maintenance or periodic checks that may be required. Safety precautions and battery disposal procedures to be followed in case of accidental activation of the battery shall also be explained. These instructions, prepared by the manufacturer, shall in no way conflict with the requirements of this specification, and a sample copy of these instructions shall be submitted to the Naval Ordnance Systems Command or its appointed agency prior to submission of the preproduction samples. A contractor may amend these instructions at a later date only with the written approval of the Naval Ordnance Systems Command or its appointed agency.

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3.3 Definitions

3.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. Amplitude may refer to displacement or acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units (g). The frequency is given in cycles per second (cps).

3.3.2 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be defined as a buildup of 90 percent of peak force in less then 0.010 second. Shock is described (a) as a pulse in gravitational units, duration in milliseconds (ms), and some description of the shock wave form, or (b) in terms of test conditions outlined in Specification MIL-S-901.

3.3.3 Vibration and shock axes. Vibration and shock axes are imaginary straight lines passing through the battery, along which lines the battery is being vibrated or shocked. The axes of the battery are:

(a) Longitudinal (axial)
(b) Perpendicular and transverse to (a)
(c) Mutually perpendicular to (a) and (b)

3.3.4 Nonoperating condition. The nonoperating condition of the battery is that condition when the battery is packaged for delivery in accordance with BUWEPS LD 495966 and LD 495967 during transportation, storage, and handling.

3.3.5 Ready, nonoperating condition. The ready nonoperating condition of the battery is that condition under which the battery is transported or handled in an unpackaged state, before or after assembly into a weapon.

3.3.6 Come-up time. Come-up time is the elapsed time, expressed in seconds, between the instant that actuation power (3.4.9) is applied and the instant that battery voltage reaches a specified value under a specified load.

3.4 Performance requirements and product characteristics. The battery shall meet the following performance requirements and product characteristics:

3.4.1 Battery assembly. The complete battery assembly shall conform strictly to the mounting and mating dimensions of BUWEPS Dwg 1803669 and shall fit within the cylindrical area shown on that drawing.

3.4.2 Silver. The amount of silver contained in one battery shall be 700 troy ounces minimum.

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3.4.3 Active material. The total silver content of all the active material, such as argentic oxide, argentous oxide, and powdered silver, in all cathodes of the propulsion and auxiliary cells shall be not less than 560 troy ounces of silver. The total silver content of the structural components in the entire battery, such as anodes and cathode grids, plate straps, terminals, and intercell connectors, shall be not less than 140 troy ounces of silver. The minimum total silver requirement for active material, 560 troy ounces, and for structural components, 140 troy ounces, shall be met independently. Any proposed excess in either of these requirements shall not be permitted to reduce the minimum requirements of the other. Only silver shall be used for structural material within the envelope of the battery cells.

3.4.3.1 Composition of active material. The active cathode material shall be composed of a minimum of 60 weight percent argentic oxide and a maximum of 10 weight percent finely divided metallic silver.

3.4.4 Electrolyte. At the time of battery manufacture, electrolyte of a type and in an amount to properly actuate the battery shall be assembled within a sealed chamber integral with the battery actuation system.

3.4.4.1 Electrolyte leakage, nonactivated. Prior to activation of the battery, there shall be no electrolyte leakage when the battery is inverted in any position.

3.4.4.2 Electrolyte leakage, activated. Once the battery has been activated, there shall be no electrolyte external to the confines of the cells proper, the electrolyte tank, and associated manifolding system during the minimum required discharge period when the battery is tilted 45 degrees in any direction from the upright position.

3.4.4.3 Electrolyte release mechanism. The system used to release the electrolyte from its storage container shall employ a contact type mechanism which punctures the seal on the container. Systems which rely solely upon a pressure buildup within the electrolyte container to rupture the seal are not acceptable.

3.4.5 Gas storage system. The battery activation system shall not employ a pressurized gas storage system for moving the electrolyte into the battery cells.

3.4.6 Base cell material. The base cell material (current collector) of the anodes and cathodes shall be either solid foil or grid mesh. Grid mesh design shall have a minimum of 300 openings to the square inch.

3.4.7 Weight. The weight of the assembled battery, ready for use, shall be 265 plus 10 minus 20 pounds.

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3.4.8 Heater. The electrolyte storage container shall be equipped with thermostatically controlled immersion type heating elements. The heater elements shall be capable of raising the temperature of the stored electrolyte to 75 plus or minus 15 degrees Fahrenheit (F) from an initial temperature of minus 20 plus or minus 2 degrees F within a 12-hour period while exposed to an ambient temperature of minus 20 plus or minus 2 degrees F.

3.4.8.1 Heater power. The battery heater circuit elements shall draw a maximum of 3.5 amperes root mean square (rms) at 115 plus or minus 5 volts, 60 cycles alternating current (ac).

3.4.9 Actuation power. The power requirements for equipment needed to empty the electrolyte into the elements shall not exceed 15 amperes rms and 15 volts rms, 60 cps.

3.4.9.1 Duration. The actuation power shall be available for 50 milliseconds maximum.

3.4.10 Discharge characteristics, series-connected. When operating under the loads specified in Figure 1, the voltage of the series-connected battery shall meet the requirements of Figure 1 for a period of 10.5 minutes following activation.

3.4.10.1 Individual cell voltages. The compliance of individual cells with this specification shall be determined by using the proper multipliers, making allowance for losses through the various intercell connectors.

3.4.11 Auxiliary tap. A separate 26 volt auxiliary load section shall have a load voltage of Figure 1, when the battery is operating under load conditions specified.

3.4.12 Discharge characteristics, parallel-connected. When operating under the loads specified in Figure 2, the voltage of the parallel-connected battery shall meet the requirements of Figure 2 for a period of 36 minutes following activation.

3.4.12.1 Auxiliary voltage, parallel. The auxiliary voltage shall be as in Figure 2, when the battery is operating with loads applied as specified in Figure 2.

3.4.13 Center of gravity. The center of gravity of the battery shall be as described on BUWEPS Dwg 1803669.

3.4.14 Shelf storage life. The complete battery shall meet the performance requirements and product characteristics of this specification at any time within a period of five years under the service conditions specified in 3.3.4 and 3.3.5. Temperatures during storage shall not exceed 110 degrees F.

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3.5 Environmental requirements

3.5.1 Vibration, ready nonoperating condition (3.3.5). The battery shall withstand vibration testing in accordance with 4.5.1.1 in each of the axes of 3.3.3, and thereafter shall meet the requirements of 3.4.10.

3.5.2 Shock, ready nonoperating condition (3.3.5). The battery shall withstand the shock tests of 4.5.1.2 on both sides, bottom, and forward end, and thereafter shall meet the requirements of 3.4.10.

3.6 Cleanup. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter.

3.7 Workmanship. The battery, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to cleanliness and neatness of manufacturing operations; the elimination of burrs, wire strands, or projections on electrode surfaces; adequate insulation of electrodes and intercell connectors; and the removal of all particles and burrs from the activation system.

The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of Battery Mk 46 Mod 1 shall be classified as follows:

a. Preproduction inspection (paragraph 4.5)
b. Acceptance inspection (paragraph 4.4)
c. Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Acceptance inspection sampling

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4.3.1.1 Sampling for destructive tests. Unless otherwise specified, a sample of cells shall be selected in accordance with MIL-STD-414, Inspection Level I (minimum sample size of seven cells) and subjected to the tests of 4.4.5.1 through 4.4.5.5.1. Acceptance or rejection of the cell lot shall be in accordance with MIL-STD-414, Table B-1, AQL 1.0 percent, AQL value for lower specification limit only.

4.3.1.2 Sample cells. Ten cells representing propulsion and eight cells representing auxiliary sections shall be submitted with preproduction batteries to the specified testing activity.

4.3.1.3 Production sample cells. Ten cells representing propulsion and eight cells representing auxiliary sections shall be randomly selected from cells which represent 75 percent of any contract quantity of batteries to be delivered. Those cells shall be analyzed for silver content (3.4.2 and 3.4.3).

4.3.1.4 Silver analysis. Unless otherwise specified in the contract or order, the sample cells of 4.3.1.3 shall be submitted to the Government for analysis.

4.3.1.5 Cell material. A quantity of 20 anodes and 15 cathodes representing propulsion cells, including separator material of each battery to be delivered, shall be packaged and shipped in each shipping container (see 5.2.1).

4.3.2 Preproduction sample. A preproduction sample of two Batteries Mk 46 Mod 1, manufactured in accordance with 3.2.2.1, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of Battery Mk 46 Mod 1 by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk. Preproduction samples accepted are over and above the quantity specified in the contract or purchase order.

4.3.3 Periodic production sampling. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.4.4 through 4.4.4.8. These samples will be subjected to the tests of 4.5 at an activity designated by the procuring activity. Periodic production samples accepted are over and above the quantity specified in the contract or purchase order. Sampling shall be as follows:

Step 1. Upon initiation of production randomly select one unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

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Step 2. Randomly select one unit from each group of 20 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 3.

Step 3. Randomly select one unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 4.

Step 4. Randomly select one unit from each group of 40 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 5.

Step 5. Randomly select one unit from each group of 50 units. Continue with Step 5 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.4 Acceptance inspection. Except where otherwise specified, the following tests and examinations are to be accomplished on each battery being submitted for acceptance under contract. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the battery shall be subjected to acceptance tests under the following conditions:

4.4.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C).

4.4.1.2 Humidity: Room ambient to 95 percent relative maximum.

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

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4.4.3 Test equipment. The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.4.3.1 Load. The load shall be of sufficient rating to test individual cells connected in series or parallel under the load conditions specified in Figures 1 and 2, and shall be capable of being adjusted to maintain tolerance throughout the discharge test time specified.

4.4.3.2 Voltmeters. Voltmeters shall have at least 1000 ohms per volt, and shall be accurate within 2 percent of voltage being measured.

4.4.3.3 Ammeters. Ammeters shall be accurate within 2 percent of value of current being measured.

4.4.3.4 Thermometers. Thermometers shall have sufficient range to record from minus 20 degrees to plus 130 degrees F and shall be accurate within 2 degrees.

4.4.3.5 Timing devices:

4.4.3.5.1 Short time interval. A timing device capable of measuring from 0 to 1 minute, in 1/2 second intervals, and accurate within 1 percent of time being measured.

4.4.3.5.2 Long time interval. A timing device capable of measuring from 0 to 30 minutes, in 1/2 minute intervals, and accurate within 1 percent of time being measured.

4.4.4 Test procedures

4.4.4.1 Preparation for tests. Prior to the performance of any tests, the battery or cells, ready for activation, shall be stored unprotected by insulation for 24 hours at the temperature and humidity conditions under which they are to be tested.

4.4.4.1.1 Unless otherwise specified, the temperature of each battery or cell shall, at the start of the test, be between 70 and 80 degrees F, and room temperature shall be between 65 and 95 degrees F for the tests. No external cooling shall be provided.

4.4.4.2 Internal shorts. The resistance between the plates or any individual cell in the dry condition shall be not less than 1000 ohms.

4.4.4.3 Polarity. The polarity of each battery shall be compatible with the polarity established for the battery, as ascertained by inspection during manufacture or by visual inspection after assembly.

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4.4.4.4 Center of gravity. The center of gravity of the battery shall be verified for conformance with the requirements of BUWEPS Dwg 1803669.

4.4.4.5 Weight. The weight of the battery, filled with electrolyte and ready for activation, shall be 265 plus 10 minus 20 pounds.

4.4.4.6 Leakage. The battery, when filled and unactivated, shall not leak electrolyte.

4.4.4.6.1 Angle of operation. There shall be no leakage of electrolyte during the first 9 minutes of normal discharge of a series-connected battery, or the first 30 minutes of normal discharge of a parallel-connected battery, when the battery is tilted at any angle between 0 and 45 degrees from an upright position.

4.4.4.7 Continuity, actuating circuit. Using a blasting galvanometer or equivalent series-type ohmmeter designed for testing blasting caps or other low-energy firing circuits, check the continuity of the battery actuating circuits. (Determination of continuity by means of conventional ohmmeters shall not be attempted.)

4.4.4.8 Heaters. With 115 plus or minus 5 volts, 60 cycles ac applied to the terminals of the heater circuit and with the thermostat closed, the current of the heaters shall not exceed 3.5 amperes. With the temperature decreasing, the thermostat shall close, and with the temperature increasing, the thermostat will open at a temperature required by the manufacturer to maintain proper operating characteristics of the battery.

4.4.5 The following tests shall be performed on sample cells selected in accordance with 4.3.1.1. Two cells of the sample shall be used to verify the appropriate parallel connections requirement (see 4.4.5.3 and 4.4.5.4). The remaining cells of the sample shall be used to verify the appropriate series connection requirement (see 4.4.5.2 and 4.4.5.4). When a separate cell group is used for the auxiliary section, a sample of these cells shall also be selected in accordance with 4.3.1.1 and tested to verify the appropriate capacity requirements (see 4.4.5.4). Appropriate paragraph selection shall be based on the method and type of cells used in supplying the auxiliary power.

4.4.5.1 Actuation power. A source of actuation power, as specified in 3.4.9, shall be applied to the actuating terminals of the battery. The actuating voltage shall be measured and plotted against time. Battery actuation shall be completed within 50 ms or less.

4.4.5.2 Discharge test, series connected. The unit cell shall be capable of being discharged as specified in 3.4.10. A continuous record of voltage versus time shall be made during the first 15 seconds of discharge.

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Recorded readings shall be made at 15 second intervals during the remainder of the first minute of discharge, and then at 1/2 minute intervals during the remainder of the discharge test. The voltage recordings shall be as specified in 3.4.10.

4.4.5.3 Discharge test, parallel connected. The unit cell shall be capable of being discharged as specified in 3.4.12. A continuous record of voltage versus time shall be made during the first 15 seconds of discharge. Recorded readings shall be made at 15 second intervals during the remainder of the first minute of discharge, and then at one minute intervals during the remainder of the discharge test. The voltage recordings shall be as specified in 3.4.12.

4.4.5.4 Discharge test, auxiliary section. The auxiliary 26 volt section shall supply the voltages specified in Figures 1 and 2, when the section is discharged under the load conditions specified in those figures. A continuous record of voltage versus time shall be made during the first 15 seconds of discharge time. Recorded readings shall be made at 15 second intervals during the remainder of the first minute of discharge, and then at one minute intervals for parallel connected batteries or 1/2 minute intervals for series connected batteries during the remainder of the discharge test. The voltage recordings shall be as specified in Figures 1 and 2.

4.4.5.5 Come-up time. The battery shall be within the voltage tolerances of Figure 1 and Figure 2 during the time shown after activating power is applied.

4.4.5.5.1 Come-up time, individual cells. The voltages listed in 4.4.5.2, 4.4.5.3, 4.4.5.4, and 4.4.5.5 are for the complete battery. If any portion of a complete battery is used to prove tests, the proper multiplying factors and inspections and test tolerances must be used to ensure that the final requirements are met. Individual cells, which have been hand-filled with electrolyte, shall not be used to test the conformance of the battery or individual cell to the voltage requirements of Figure 1 and Figure 2 during the first 15 seconds of discharge.

4.4.5.6 Quantitative analysis of active cathode material. The method for performing the quantitative analysis of the active cathode material shall be in accordance with the procedures outlined in Naval Ordnance Laboratory, White Oak, Maryland, Report NOLTR 64-214 tilted "A Method for the Quantitative Analysis of the Silver Oxide Cathode."

4.4.6 Visual and dimensional examination. Each battery shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.7 Packaging and packing. The packaged battery and the test material packages shall be examined to ensure that they comply fully with the requirements of Section 5.

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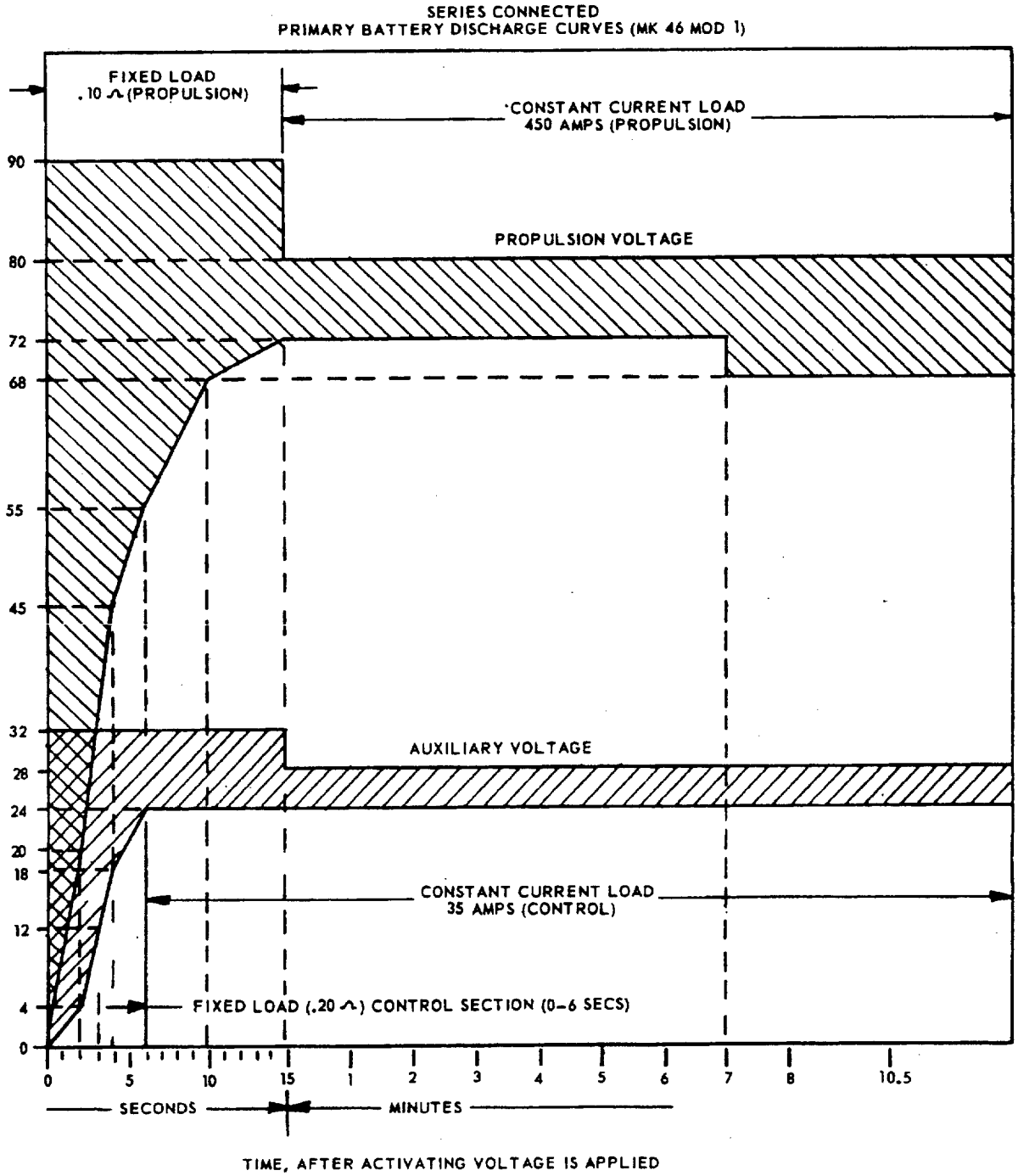


FIGURE 1

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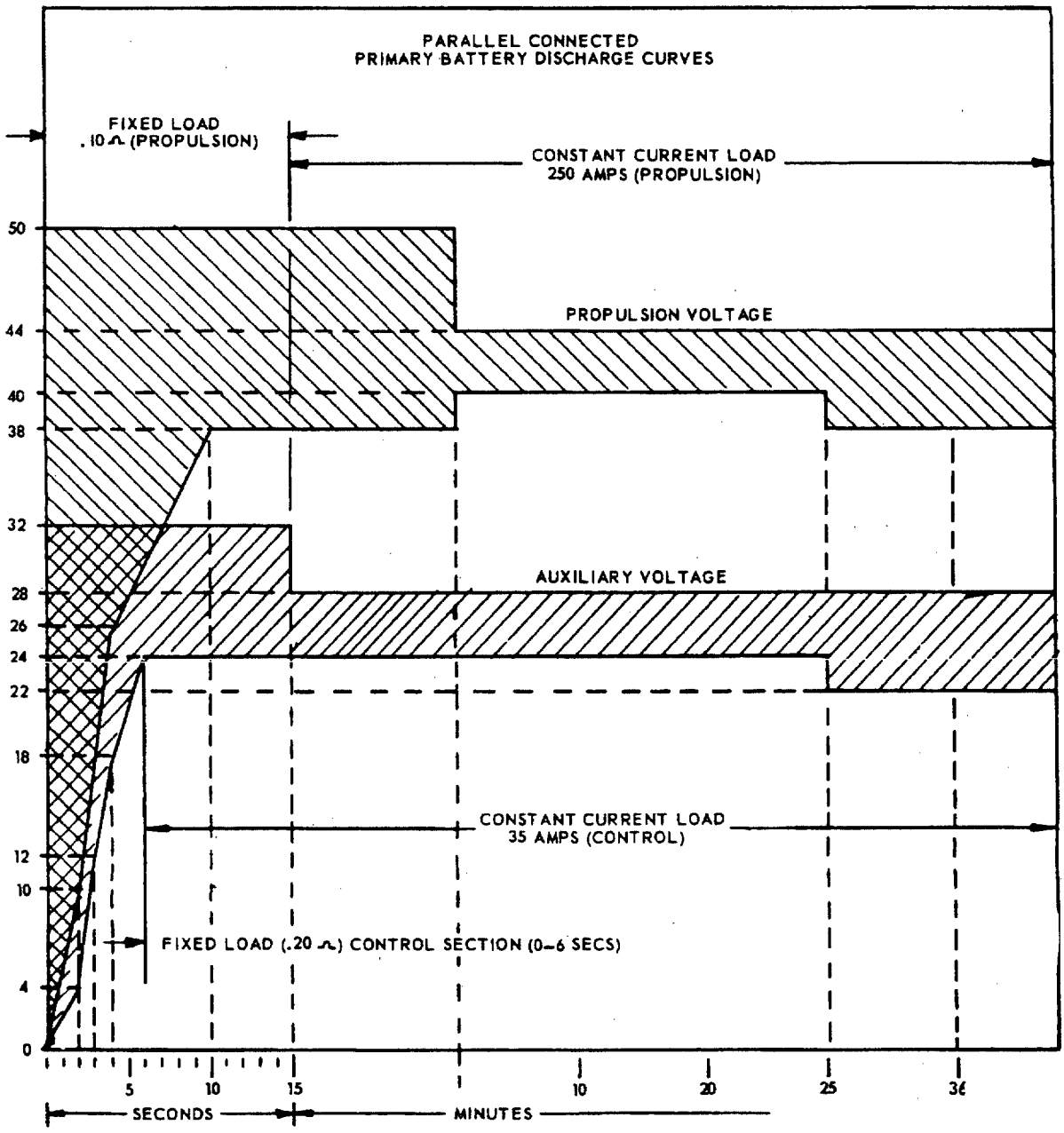


FIGURE 2

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4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples specified in 4.3.2 and 4.3.3, after satisfactorily passing the acceptance inspections of 4.4.4 through 4.4.4.8, shall be subjected to the following tests:

4.5.1 Vibration and shock, ready nonoperating condition. Vibration and shock tests will normally be conducted by a Government facility. Tests shall simulate the conditions for normal environment as specified and defined in 3.3. It shall be the Government's prerogative to decide the tests, methods, and equipment used. The tests conducted shall not exceed those specified herein.

4.5.1.1 Vibration. The battery shall be vibration tested in accordance with Schedule I and thereafter shall meet all the requirements of 3.5.1.

SCHEDULE I (VIBRATION)

Frequency Range	Amplitude	Minimum Test Duration		
		Room Temp.	-20° F	+130° F
10 - 30 cps	0.060 ± 0.006 in.	60 min.	15 min.	15 min.
31 - 60 cps	3g ± 0.3g	60 min.	15 min.	15 min.
61 - 100 cps	3g ± 0.3g	60 min.	15 min.	15 min.

4.5.1.2 Shock. The battery shall be shock tested in accordance with Schedule II and thereafter shall meet all the requirements of 3.5.2.

SCHEDULE II (SHOCK)

Shock Pulses in Each Direction	Peak Magnitude	Minimum Pulse Duration	Pulse Shape
3	60g ± 6g	8 ms	Approximating a one-half sine wave

4.6 Acceptance criteria. Failure of the battery to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Battery

5.1.1 Preservation and packaging. Each battery shall be preserved and packaged in accordance with the drawings, specifications, and publications listed in BUWEPS LD 495966.

5.1.2 Packing, Level A. Batteries packaged in accordance with 5.1.1 shall be packed in accordance with the drawings, specifications, and publications listed in BUWEPS LD 495967.

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5.1.3 Marking. Each battery container shall be marked in accordance with the drawings, specifications, and publications listed in BUWEPS LD 495967.

5.2 Test Material

5.2.1 Packaging. The cell material (4.3.1.5) shall be packed in five separate packages, each containing three positive and four negative electrodes and applicable separator material. Each test material set shall be wrapped in chemically neutral wrapping paper (Specification MIL-P-17667) and then wrapped in 0.5 inch-thick cellulose cushioning material (Specification PPP-C-843, Type II, Class B), secured with tape (Specification PPP-T-60, Class III), and placed in a snug fitting heat sealed bag (Specification MIL-B-117, Type I, Class D). Tape the packaged test material to end of box assembly, BUWEPS LD 495852, after placing box in metal drum, using 3/4 inch minimum width tape (Specification PPP-T-60, Class III).

5.2.2 Marking. Each of the five test material packages (5.2.1) shall be marked as follows:

"QUALITY ASSURANCE MATERIAL,
DO NOT OPEN WHEN THIS BATTERY IS PLACED IN SERVICE.
PROVIDE OUTER PACKAGING AND FORWARD TO:

QUALITY EVALUATION LABORATORY
NAVAL TORPEDO STATION
KEYPORT, WASHINGTON 98345

MFG'S NAME _____

DATE OF MFG _____

CONTRACT NO. _____

SERIAL NO. OF BATTERY REPRESENTED _____ "

6. NOTES

6.1 Intended use. The battery is intended to provide electric power for torpedo propulsion and control.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Assigned activity for preproduction and periodic production tests.

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(c) Any exceptions to this specification.

(d) Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Naval Ordnance Systems Command when testing is accomplished at other than a Government activity.

6.3 The quantity of material required to produce preproduction and periodic production batteries or cells used for test purposes defined in this specification, should be considered as additional material required to produce a production quantity of batteries for delivery under any contract or order.

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